

Advanced Machine Learning

DATA 642

Lecture 3 — Constrained Optimization

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth

Constrained Optimization and Lagrange Multipliers

In unconstrained optimization, we considered the problem of solving for the minimum of a function

$$\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) \tag{1}$$

where $J : \mathbb{R}^l \rightarrow \mathbb{R}$.

In constrained, we have additional constraints. That is, for real-valued functions $g_i : \mathbb{R}^l \rightarrow \mathbb{R}$ for $i = 1, \dots, m$, we consider the constrained optimization problem

$$\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) \tag{2}$$

subject to $g_i(\boldsymbol{\theta}) \leq 0$ for all $i = 1, \dots, m$.

It is worth pointing out that the functions J and g_i could be non-convex in general, and we will consider the convex case in the next section.

Constrained optimization problems occur when $S \subset \mathbb{R}^l$ can be defined implicitly by a set of constraints:

$$S = \{\boldsymbol{\theta} \mid g_i(\boldsymbol{\theta}) = 0; g_i(\boldsymbol{\theta}) \leq 0\}, \quad (3)$$

for any $i = 1 \dots m$.

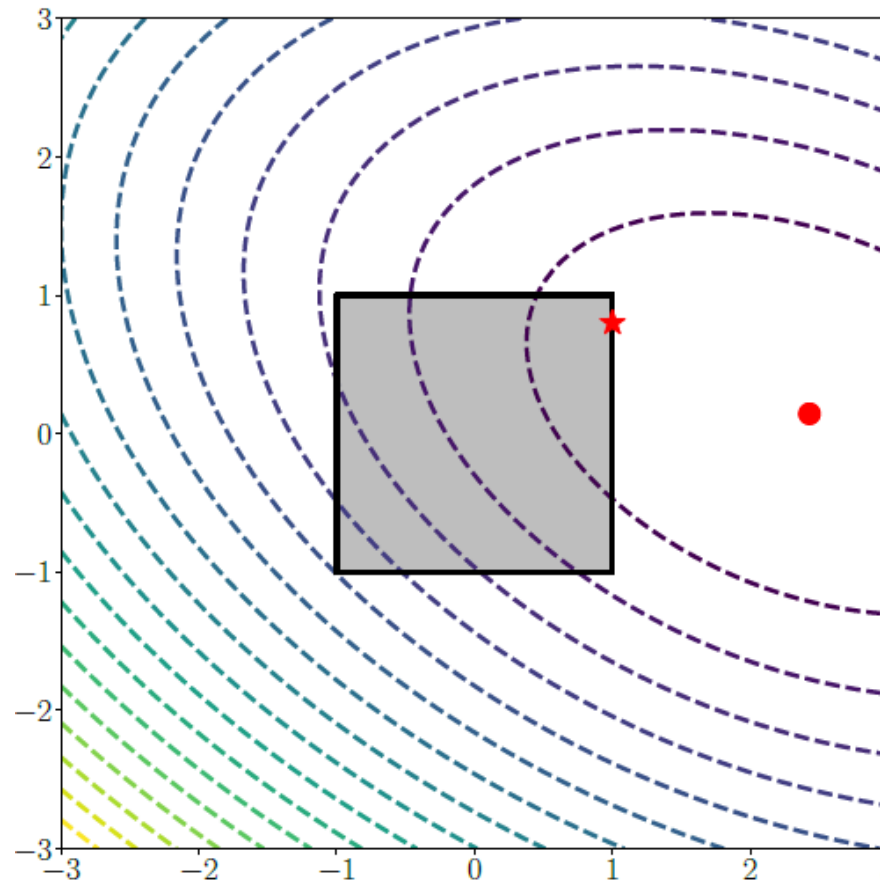


Illustration of constrained optimization. The unconstrained problem (indicated by the contour lines) has a minimum on the right side (indicated by the circle). The box constraints $-1 \leq \theta_1 \leq 1$ and $-1 \leq \theta_2 \leq 1$ require that the optimal solution is within the box, resulting in an optimal value indicated by the star.

Understanding the Primal Problem

- The primal problem corresponds to the original constrained optimization problem that we aim to solve.
- Mathematically, the primal problem can be expressed as:

$$\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) \tag{4}$$

subject to $g_i(\boldsymbol{\theta}) \leq 0, \quad i = 1, \dots, m$

- The primal problem provides the optimal value of the objective function and the corresponding values of the original variables $\boldsymbol{\theta}$.
- Solving the primal problem directly may not always be straightforward, especially when dealing with complex constraints or non-convex objective functions.

Converting a constrained optimization problem into an unconstrained one is important for several reasons:

1. **Simplicity:** Unconstrained optimization problems are often simpler to solve computationally compared to constrained ones. Many optimization algorithms are designed to handle unconstrained problems efficiently, making the optimization process faster and more straightforward.
2. **Algorithmic Compatibility:** Some optimization algorithms are specifically designed for unconstrained optimization. By converting a constrained problem into an unconstrained one, you can leverage a wider range of optimization algorithms, including those that are highly efficient and widely used.
3. **Theoretical Analysis:** Unconstrained optimization problems often have well-established theoretical properties and convergence guarantees, which can facilitate theoretical analysis and provide insights into the behavior of optimization algorithms.
4. **Solving Complex Problems:** Converting a constrained problem into an unconstrained one may enable the use of advanced optimization techniques that are tailored for unconstrained problems. This can be particularly useful for solving complex optimization problems with nonlinear or non-convex objectives and constraints.

To convert the constrained minimization problem to an unconstrained we introduce the Lagrangian function which is defined as follows

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = J(\boldsymbol{\theta}) - \sum_{i=1}^m \lambda_i g_i(\boldsymbol{\theta}) = J(\boldsymbol{\theta}) - \boldsymbol{\lambda}^\top \mathbf{g}(\boldsymbol{\theta}), \quad (5)$$

where in the last line we have concatenated all constraints $g_i(\boldsymbol{\theta})$ into a vector $\mathbf{g}(\boldsymbol{\theta})$, and all the Lagrange multipliers into a vector $\boldsymbol{\lambda} \in \mathbb{R}^m$.

- The task is to identify points that obey the first-order conditions of the Lagrangian; i.e.,

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = \mathbf{0}, \quad \nabla_{\boldsymbol{\lambda}} \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = \mathbf{0}. \quad (6)$$

- Lagrange multiplier can be interpreted as the rate of change of $J(\boldsymbol{\theta})$ w.r.t. changes in the corresponding constraint function.

Understanding the Lagrange Dual Problem

The purpose of the Lagrange dual problem is multifaceted, with several key objectives and implications:

1. **Bounding the Optimal Solution:** One of the primary purposes of the dual problem is to provide a bound on the optimal value of the original constrained optimization problem. By solving the dual problem, we obtain a lower bound on the optimal objective function value, known as the dual optimal value. This bound can help us assess the quality of solutions obtained through various optimization algorithms and guide the search for better solutions.
2. **Insights into the Primal Problem:** Solving the dual problem often provides valuable insights into the solutions of the primal problem. The optimal dual variables (Lagrange multipliers) correspond to the sensitivity of the primal problem's constraints. By analyzing these dual variables, we can gain a deeper understanding of how changes in the constraints affect the primal problem's objective function and optimal solution.
3. **Efficient Computational Approaches:** In some cases, solving the dual problem can lead to more efficient computational approaches for solving the primal problem. Dual optimization algorithms, such as dual decomposition or cutting-plane methods, exploit the structure of the dual problem to achieve faster convergence and lower computational complexity compared to primal optimization algorithms.

The Lagrange dual problem involves maximizing the Lagrangian function $\mathcal{L}$ with respect to the Lagrange multipliers λ , while minimizing with respect to the original variables θ .

The associated Lagrangian dual problem is given by

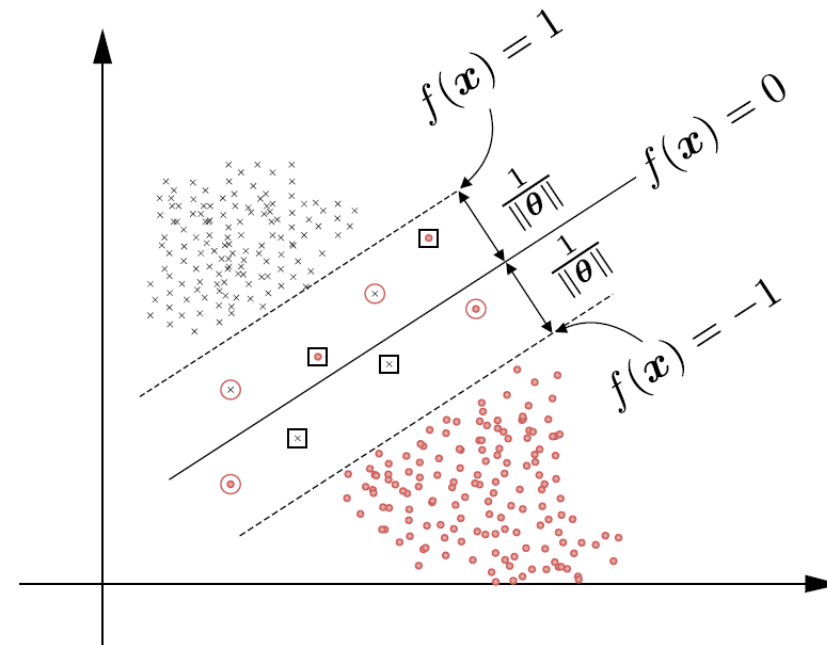
$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^m} D(\boldsymbol{\lambda}) \quad \text{subject to} \quad \boldsymbol{\lambda} \geq 0 \quad (7)$$

where $\boldsymbol{\lambda}$ are the dual variables and $D(\boldsymbol{\lambda}) = \min_{\boldsymbol{\theta} \in \mathbb{R}^d} \mathcal{L}(\boldsymbol{\theta}; \boldsymbol{\lambda})$.

An example

- SVM optimization seeks to minimize the objective function while subject to certain constraints:

$$\begin{aligned} & \underset{\boldsymbol{\theta} \in \mathbb{R}^l}{\text{minimize}} && \frac{\|\boldsymbol{\theta}\|^2}{2} + C \sum_{n=1}^N \xi_n \\ & \text{subject to} && y_n(\mathbf{x}_n \boldsymbol{\theta} + b) \geq 1 - \xi_n; \quad \xi_n \geq 0 \end{aligned} \tag{8}$$



- By introducing Lagrange multipliers and forming the Lagrangian, we can derive the Lagrange dual problem for SVM optimization. Solving the Lagrange dual problem allows us to obtain insights into the SVM solution and can lead to more efficient computational approaches.